Depletion of Solar Wind Plasma Near a Planetary Boundary

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A mathematical model is presented that describes the squeezing of solar wind plasma out along interplanetary magnetic field lines in the region between the bow shock and the effective planetary boundary (in the case of the earth, the magnetopause). In the absence of local magnetic merging the squeezing process should create a 'depletion layer,' a region of very low plasma density just outside the magnetopause. Numerical solutions are obtained for the dimensionless magnetohydrodynamic equations describing this depletion process for the case where the solar wind magnetic field is perpendicular to the solar wind flow direction. For the case of the earth with a magnetopause standoff distance of 10 R_E, the theory predicts that the density should be reduced by a factor ≥ 2 in a layer about 700–1300 km thick if M_A, the Alfvén Mach number in the solar wind, is equal to 8. The layer thickness should vary as M_A^{-2} and should be approximately uniform for a large area of the magnetopause around the subsolar point. Our computed layer thicknesses are somewhat smaller than those derived from Lees' axisymmetric model. Depletion layers should develop fully only where magnetic merging is locally unimportant. Scaling of the model calculations to Venus and Mars suggests layer thicknesses about $\frac{1}{10}$ and $\frac{1}{18}$ those of the earth, respectively, neglecting diffusion and ionospheric effects.

INTRODUCTION

It is well known that many plasma flow features of the solar wind–geomagnetic interaction can be obtained from the continuum equations of magnetohydrodynamics. It has been established [*Spreiter et al.*, 1966; *Spreiter and Alksne*, 1968, 1969] that simple gasdynamics is sufficient to reproduce the broad characteristics of plasma flow around the magnetosphere.

Starting from the gasdynamic flow fields computed by Spreiter and co-workers, *Alksne* [1967] used the assumption of frozen-in flux ($\mathbf{E} + \mathbf{v} \times \mathbf{B}/c = 0$) to derive magnetic field patterns in the magnetosheath for several assumed orientations of the magnetic field in the upstream solar wind. Let region R be the portion of the day side magnetosheath within 10 R_E of the symmetry axis, excluding a boundary layer 0.5 R_E thick just outside the magnetopause; for this region the calculations of *Alksne* [1967] imply that

$$\langle B^2/8\pi\rangle/\langle p_g + \tfrac{1}{2}\rho v^2\rangle \sim (M_A(SW)/3)^{-2}$$

where p_g is the gas pressure and $M_A(SW)$ is the Alfvén Mach number in the upstream solar wind. Since $M_A(SW)$ is usually $\gtrsim 10$, it is reasonable to consider magnetic stress a small perturbation in region R unless $M_A(SW)$ is unusually small or high accuracy is required. Neglect of magnetic stress in region R becomes exactly valid as $M_A(SW) \to \infty$. For any finite value of $M_A(SW)$, however, *Alksne*'s [1967] computed magnetic field goes to infinity at the magnetopause, which suggests that there is always a small region of the magnetosheath near the magnetopause where magnetic stress strongly affects gas flow. It is this region, the depletion layer, that we model in the present paper.

The straightforwardly rigorous approach of evaluating magnetic effects on magnetosheath flow would be to solve the full three-dimensional MHD equations with a shock, an enormously difficult task with present computing techniques. Our more modest procedure is to treat the thin layer outside the magnetopause in detail, including magnetic stresses there in an approximate way. We assume that the gasdynamic calculations of Spreiter and co-workers are approximately valid outside this layer and use their results to supply a crucial pressure boundary condition.

The present work constitutes an extension and an elaboration of the paper by *Lees* [1964], who considered magnetic stress and obtained a magnetohydrodynamic solution for the solar wind–geomagnetic interaction for a restricted region, the stagnation streamline. He also assumed an axisymmetric magnetosheath velocity field. Although this assumption was somewhat unrealistic, his results were extremely important, since they pointed out that inclusion of a transverse component of the interplanetary field in the plasma flow analysis reversed the computed gradient of the plasma density along the stagnation streamline; he predicted a density minimum at the stagnation point, where gasdynamics predicted a density maximum.

In this paper we apply the MHD approach to plasma flow near the magnetopause, performing the calculation for regions off the stagnation streamline and for nonaxisymmetric flow. We also examine the dependence of our results on solar wind parameters. The treatment is aimed primarily at the earth's magnetosheath, but scaling to other cases (particularly Mars and Venus) is discussed.

FORMULATION OF THE MODEL

Our depletion model describes the process whereby portions of interplanetary flux tubes, portions that lie in the magnetosheath near the day side magnetopause, become depleted of plasma. By depleted we mean that the density and the gas pressure drop well below the values achieved just after passage through the bow shock, while the total pressure (gas and magnetic) does not drop below its postshock value. Two processes deplete the tubes. By deflecting plasma around the magnetosphere the bow shock kicks plasma out along the field lines away from the nose of the magnetosphere, and the compressional stress exerted on magnetosheath flux tubes near the nose tends to squeeze plasma out along the field lines, further depleting them. Idealized sketches of this depletion process are shown in Figure 1. In particular we are illustrating the time evolution of a single thin flux tube embedded in the solar wind incident upon the magnetosheath. In our depletion model we consider the frozen-in interplanetary field to be convected

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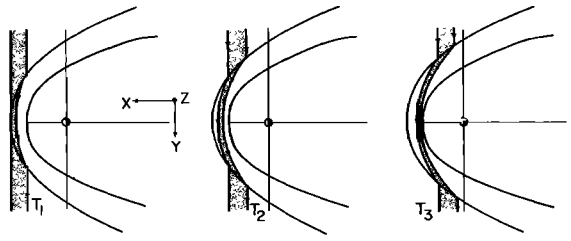

Fig. 1. Idealized sketches of the squeezing process at successive times T_1, T_2, and T_3. As plasma convects around the magnetosphere, flux tubes become draped around the nose of the magnetosphere.

tangential to the magnetopause, approximate the magnetosphere as a blunt-nosed object of revolution but do not assume the magnetosheath flow field to be axisymmetric, and assume that the magnetic field B_{SW} in the incident solar wind is perpendicular to the flow direction.

There are two reference frames. The XYZ system has its X axis pointing opposite to $\mathbf{v}_{SW}$, the velocity of the incident solar wind (i.e., the X axis points approximately toward the sun); the Y axis is parallel to $\mathbf{B}_{SW}$, and the Z axis is arranged to form an orthogonal right-hand system. The origin of the coordinate system is at the earth, but this system is not necessarily either solar magnetospheric or solar ecliptic, since its orientation depends on $\mathbf{B}_{SW}$. The xyz system is curvilinear and time dependent; its y axis is everywhere parallel to the magnetosheath magnetic field, and field lines are curves of constant x and z. The xyz system $\mathbf{E} \times \mathbf{B}$ drifts with the plasma, allowing us to follow the motion of a tube of magnetic flux through the magnetosheath. The $y = 0$ surface coincides with the $Y = 0$ plane, which contains the earth and is perpendicular to $\mathbf{B}_{SW}$. The y coordinates gives the distance from the $Y = 0$ plane as measured along a field line. It never becomes necessary to define the x and z coordinates completely.

As the flux tube is compressed, the plasma inside will tend to squeeze toward the ends. (We show in Appendix A that the cross-field diffusion does not substantially compete with the squeezing process.) This redistribution of plasma has the net effect of leaving a region around the point of maximum compression depleted of plasma. Evidently, the longer a given flux tube is compressed, the more depleted it becomes. There is a stagnation point in the magnetosheath flow where the X axis intersects the magnetopause (see Figure 2). Flux tubes that in their traversal of the magnetosheath come close to the magnetopause also come close to the stagnation point and thus experience low flow speeds. The nearer a flux tube trajectory comes to the stagnation point, the longer the tube resides in the magnetosheath and the closer it comes to the magnetopause. Thus the most depleted flux tubes lie closest to the magnetopause.

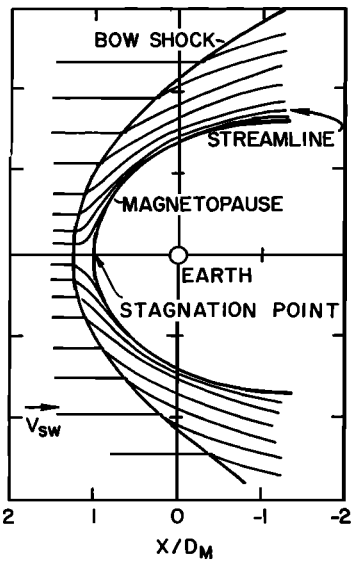

Fig. 2. Flow streamlines in the magnetosheath, adapted from *Spreiter et al.* [1968].

The plasma density and the cross-sectional area of the flux tubes decrease as depletion time increases. The ratio of the density at time zero to the density at some depletion time t is always greater than one, and we call this ratio the depletion factor, since it is a measure of the amount of depletion for a given depletion time. Our method of computing the depletion factor is based on the fact that one can get a useful approximate solution for flow within a flux tube without knowing in detail the trajectory of the tube through the magnetosheath. Our approach is therefore the following: (1) to calculate the motion of plasma within a given flux tube thus computing the density $\rho(y, t)$ and magnetic field $B(y, t)$, (2) to calculate the motion of the intersection of the flux tube with the X-Z plane ($Y = y = 0$ plane) perpendicular to the magnetic field, specifically, to compute as a function of time, the distances of the flux tube from the magnetopause and from the stagnation streamline as measured in the X-Z plane, and (3) to combine the results of the previous two calculations to obtain ρ and B as functions of distance from the magnetopause for a given region of the magnetopause.

The next three sections describe detailed calculations of density and velocity profiles just outside the magnetopause. A brief nonrigorous discussion of the physics of the layer is contained in the section 'Simple Formula for Depletion Layer Thickness.'

BASIC EQUATIONS USED IN THE MODEL

Equations describing fluid motion along field lines. We consider a model flux tube with very small cross-sectional area a perpendicular to the y axis. The equations expressing conservation of mass and momentum in a plasma can be written

$$\partial\rho/\partial t + \nabla\cdot(\rho\mathbf{v}) = 0 \qquad (1a)$$

$$(\partial/\partial t)(\rho\mathbf{v}) + \nabla\cdot\mathsf{T} = 0 \qquad (1b)$$

where the total stress tensor is given by

$$\mathsf{T} = \rho\mathbf{v}\mathbf{v} + \mathsf{P} + (B^2/8\pi)\mathsf{I} - (\mathbf{B}\mathbf{B}/4\pi)$$

where ρ is the mass density, $\mathbf{v}$ is the flow velocity (momentum density divided by mass density), P is the particle pressure tensor, $\mathbf{B}$ is the magnetic field, and I is the identity tensor. The equations do not depend on any 'fluid approximation' but are straightforwardly valid only for a coordinate system fixed in an inertial reference frame. Also, electric field stresses exerted on charge densities are neglected, as is the momentum density of the electromagnetic field, since they are extremely small in the magnetosheath.

We now apply (1a) and (1b) to motion within a tube of magnetic flux. In order to concentrate on motion within the tube rather than motion of the tube, we choose the xyz system. The effects of noninertiality of this reference frame are relatively minor and can be compensated approximately by the proper choice of initial conditions (see Appendix D).

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Let the flux tube have a small cross-sectional area a perpendicular to $\mathbf{B}$; this area varies with y in proportion to $|\mathbf{B}|^{-1}$. Integrating equations (1a) and (1b) over the small cross-sectional area and dividing the second equation by a gives

$$\frac{\partial}{\partial t}\rho a + \frac{\partial}{\partial y}\rho a v = 0 \tag{2a}$$

$$\rho\frac{\partial v}{\partial t} + \rho v\frac{\partial v}{\partial y} = -\frac{\partial}{\partial y}p_g = -\frac{\partial}{\partial y}p_{ext} + \frac{B}{4\pi}\frac{\partial}{\partial y}B \tag{2b}$$

where v and B are the y components of the plasma velocity and the magnetic field; we have made the fluid mechanical assumption of isotropic pressure,

$$P_{ij} = p_g\delta_{ij}$$

and have defined the 'external pressure' as

$$P_{ext} = p_g + p_M = p_g + (B^2/8\pi) \tag{3a}$$

The external pressure is the sum of the particle (or gas) pressure p_g and the magnetic pressure p_M. We treat the particles as an adiabatic gas with $\gamma = \frac{5}{3}$ and thus write

$$p_{ext} = K\rho^\gamma + (B^2/8\pi) \tag{3b}$$

Also, by the definition of a,

$$B^2 = \phi^2/a^2 \tag{4}$$

a function of y and t, where ϕ is a constant, the magnetic flux in the tube.

The coefficient K is assumed to be independent of y, a nontrivial assumption because fluid at different y values encounters shocks of different strengths and thus has different entropies and different K values. However, the K values deduced from *Spreiter et al.* [1966] vary by less than 20% over the range $|y| \lesssim 10\ R_E$, and we shall apply the calculations only to smaller $|y|$ values.

If p_{ext} is specified as a function of y and t, then equations (2a), (2b), (3b), and (4) can be solved for ρ, a, v, and B. With the approximations made, the one-dimensional motion within the single thin flux tube can be determined.

We form the following dimensionless variables (denoted by asterisks):

$$\begin{aligned} \rho &= \rho^*(p_0/K)^{1/\gamma} \\ a &= a^*(\phi^2/8\pi p_0)^{1/2} \\ v &= v^*C_{A0} = v^*B_0(4\pi\rho_0)^{-1/2} \\ y &= y^*l \\ t &= t^*l(C_{A0})^{-1} \\ p_{ext} &= p_{ext}{}^*p_0 \end{aligned} \tag{5}$$

where C_{A0} and p_0 represent the Alfvén speed and the external pressure (particle plus magnetic) at $y = 0$ and $t = 0$, the time when the flux tube first encountered the bow shock and l is a scaling parameter that will be discussed later. Substitution of (5) into (2b) yields

$$\begin{aligned} &\rho^*(p_0/K)^{1/\gamma}C_{A0}{}^2(\partial v^*/\partial t^* + v^*\ \partial v^*/\partial y^*) \\ &= -p_0\ \partial p_{ext}{}^*/\partial y^* - (2p_0/a^{*3})\ \partial a^*/\partial y^* \end{aligned} \tag{6}$$

Using the relation

$$p_{ext}{}^* = \rho^{*\gamma} + a^{*-2} \tag{7}$$

derived from (3b), (4), and (5), we can eliminate ρ^* from (6):

$$\frac{\partial v^*}{\partial t^*} + \frac{1}{\xi}\left(\frac{(1-\xi)}{p_{ext}{}^* - a^{*-2}}\right)^{1/\gamma} a^{*-3}\frac{\partial a^*}{\partial y^*} + v^*\frac{\partial v^*}{\partial y^*} = \frac{-1}{2\xi}\left(\frac{(1-\xi)}{p_{ext}{}^* - a^{*-2}}\right)^{1/\gamma}\frac{\partial p_{ext}{}^*}{\partial y^*} \tag{8}$$

where we have written the ratio of magnetic pressure to external pressure at $y = 0$ and $t = 0$ as

$$\xi \equiv B^2/8\pi p_{ext}|_{y=0,t=0} \equiv p_{M0}/p_0 < 1 \tag{9}$$

Substitution of (9) and (5) into (2a) yields

$$\frac{\partial a^*}{\partial t^*} + v^*\frac{\partial a^*}{\partial y^*} + a^*\frac{(p_{ext}{}^* - a^{*-2})}{[p_{ext}{}^* + (5a^{*2})^{-1}]}\frac{\partial v^*}{\partial y^*} = \frac{-3a^*v^*}{5[p_{ext}{}^* + (5a^{*2})^{-1}]}\frac{\partial p_{ext}{}^*}{\partial y^*} \tag{10}$$

where we have substituted $\gamma = \frac{5}{3}$ and have assumed $p_{ext}{}^*$ to be independent of time. In our model of the depletion layer we choose $p_{ext}{}^*(y^*)$ to match the magnetopause pressures calculated gasdynamically by *Spreiter et al.* [1966] for the region near the subsolar point. To accomplish this match, we choose the form

$$p_{ext}{}^* = (1 + y^{*2})^{-1} \tag{11}$$

(See Appendix B for detailed justification.) Note that in (11) we have implicitly defined l as

$$l^2 = -[(2p_{ext})^{-1}]\ \partial^2 p_{ext}/\partial y^2|_{y=0}$$

Using (11), we can eliminate p_{ext} and its gradient from (10) and (8). The results are

$$\frac{\partial a^*}{\partial t^*} + v^*\frac{\partial a^*}{\partial y^*} + 5a^*\frac{(a^{*2} - 1 - y^{*2})}{(5a^{*2} + 1 + y^{*2})}\frac{\partial v^*}{\partial y^*} = \frac{6a^{*3}v^*y^*}{(1 + y^{*2})}(5a^{*2} + 1 + y^{*2})^{-1} \tag{12}$$

$$\frac{\partial v^*}{\partial t^*} + v^*\frac{\partial v^*}{\partial y^*} + (a^{*3}\xi)^{-1}\left(\frac{(1-\xi)(a^{*2} + y^{*2}a^{*2})}{a^{*2} - 1 - y^{*2}}\right)^{3/5}\frac{\partial a^*}{\partial y^*} = \frac{y^*\xi^{-1}}{(1 + y^{*2})^2}\left(\frac{(1-\xi)(a^{*2} + y^{*2}a^{*2})}{a^{*2} - 1 - y^{*2}}\right)^{3/5} \tag{13}$$

Equations (12) and (13) are the dimensionless, coupled time dependent equations that describe the motion of plasma parallel to the magnetic field in the flux tube. They are written in terms of the dimensionless area of the flux tube a^*, the dimensionless plasma velocity parallel to the field v^*, and the initial value parameter ξ given by (9).

Equations describing flux tube motion in the X-Z plane. We turn now from motion inside the flux tube to the motion of the flux tube, specifically the motion of its intersection with the $Y = 0$ plane (X-Z plane).

In the analysis of plasma flow around the magnetosphere in the X-Z plane, streamlines tend to converge or to diverge depending on pressure gradients. The perpendicular distance dl separating a particular pair of neighboring streamlines depends on the position along the streamlines. Assuming flow in the magnetosheath to be steady and the magnetic field to be frozen into the plasma (the special Ohm's law $\mathbf{E} = -\mathbf{v}\times\mathbf{B}/c$), we conserve magnetic flux by demanding that the number of field lines that cross a dl connecting a particular pair of stream-

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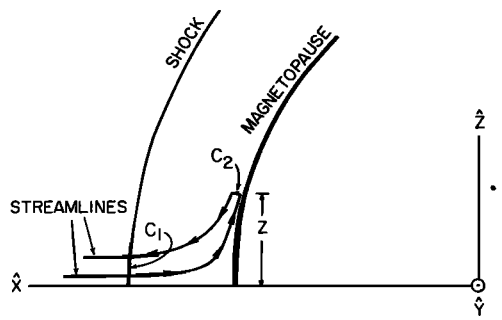

Fig. 3. A diagram illustrating the sense of the integration of (14). The distance between streamlines has been exaggerated.

lines is the same for any dl connecting that same pair of streamlines. The condition of magnetic flux conservation becomes

$$\int_{C_1} v_\perp B\, dl_1 - \int_{C_2} v_\perp B\, dl_2 = 0 \tag{14}$$

where $v_\perp$ is the velocity perpendicular to dl (see Figure 3). Along path C_1, $v_Z \approx 0$; similarly, along path C_2, $v_X \approx 0$. Thus (14) can be written

$$\int_{C_1} v_{X0} B_0 \left(\frac{\partial Z_0(Z, t)}{\partial t}\right)_Z dt - \int_{C_2} v_Z(Z, t) B(t) \left(\frac{\partial D(Z, t)}{\partial t}\right)_Z dt = 0 \tag{15a}$$

where v_{X0} and B_0 are, respectively, the X component of flow velocity and the magnetic field just postshock. The distance differential along C_1 is dZ_0, which here is written $\partial Z_0/\partial t \cdot dt$. (It turns out to be convenient to label a streamline by the time t required for a fluid element to move along that streamline from an initial height Z_0 above the X axis just postshock to a distance Z from the X axis.) Analysis of flow along the field line gives B at $y = 0$ as a function of convection time t; similarly, an analysis to be given below gives v_Z as a function of Z and t. The obvious integration variable for integration along C_2 is D, the distance from the magnetopause, but dD is conveniently rewritten $\partial D/\partial t \cdot dt$. Since t is a streamline label, the range of integration over t is the same for both integrals in (15a), and we can write

$$\int \left[v_{X_0} B_0 \left(\frac{\partial Z_0(Z, t)}{\partial t}\right)_Z - v_Z(Z, t) B(t) \left(\frac{\partial D(Z, t)}{\partial t}\right)_Z \right] dt = 0 \tag{15b}$$

We will use (15b) to find $D(Z, t)$. Inverting it will give depletion time as a function of distance D from the magnetopause, which combined with knowledge of $\rho(t)$ at $y = 0$ derived from analysis of flow along the field line, will allow calculation of density as a function of distance from the magnetopause.

We now calculate the Z coordinate of the flux tube in the X-Z plane as a function of time. We approximate the magnetosphere as a blunt-nosed body of revolution (about the earth-sun line) and assume that

$$\partial^2 p_{ext}/\partial Z^2 \approx \partial^2 p_{ext}/\partial Y^2 \tag{16}$$

Restricting the analysis to small Z, the region near the symmetry axis, we expand the external pressure in a Taylor series about $Z = 0$, where it has a maximum:

$$\partial p_{ext}/\partial Z \simeq Z|\partial^2 p_{ext}/\partial Z^2|_{Z=0} \tag{17}$$

The Z component of the momentum equation for motion of the flux tube in the magnetosheath becomes

$$d^2Z/dt^2 = Z/\tau(t)^2 \tag{18}$$

where

$$\tau(t)^{-2} = [\sigma^2/\rho(t)]|\partial^2 p_{ext}/\partial Z^2|_{Z=0} \tag{19}$$

and σ is a dimensionless factor intended to take approximate account of the anisotropy of the magnetic stress, i.e., the magnetic tension. If there were no magnetic field or if the field lines were straight near $y = 0$, then the volume force acting on fluid elements in the X-Z plane tending to push them along the magnetopause would simply be the gradient of $p_{ext} = (p_g + p_M)$; this situation would correspond to $\sigma = 1$. It is impossible to include effects of magnetic tension accurately without solving the entire three-dimensional flow problem. However, in Appendix C we overestimate the magnetic stress and derive an approximate upper limit of 1.25 for σ.

For initial conditions $Z(t = 0) = Z_0$ and $v_Z(t = 0) = v_{Z0}$ and for the case where $|d\tau(t)/dt| \ll 1$, the WKB approximation yields the following solution to (18):

$$Z(t) \simeq \left(\frac{\tau(t)}{\tau_0}\right)^{1/2} \left(\frac{Z_0}{2}\right)\left(1 + \frac{v_{Z0}\tau_0}{Z_0}\right) \exp\,[I(t)] \tag{20}$$

where $I(t) = \int_0^t dt'/\tau(t')$ and $\tau_0 = \tau(0)$. Similar to the dimensionless quantities given in (5), we define

$$D = D^* l \tag{21a}$$

and

$$\tau = \tau^* l (C_{A0})^{-1} \tag{21b}$$

Using the dimensionless quantities in (21), setting the bracketed term equal to zero in (15b), since the time integration is arbitrary, and using (20) to solve for $v_Z(t)$ yield the dimensionless depletion layer thickness

$$D^*(t^*) = S^* \int_{t^*}^{\infty} dt^{*\prime\prime} \left(\frac{\tau_0^*}{\tau^*(t^{*\prime\prime})}\right)^{1/2} \frac{a^*(t^{*\prime\prime})}{a_0^*} \cdot \exp\,[-I(t^{*\prime\prime})] \tag{22}$$

To arrive at (22), we have used (4) to eliminate the magnetic field terms and have formed the quantities

$$S^* = 2|v_{X0}^*|[1 + (v_{Z0}\tau_0/Z_0)]^{-1} \tag{23}$$

$$I(t^*) = \int_0^{t^*} dt^{*\prime}/\tau^*(t^{*\prime}) \tag{24}$$

Equations (22)–(24) give the distance from the magnetopause to the flux tube as measured in the X-Z plane at a particular convection time t^*. All quantities refer to the X-Z plane ($y = 0$), and the subscript 0 refers to time $t^* = 0$. The solutions of (22)–(24) explicitly depend on $a^*(t^*)$ and implicitly on $\rho^*(t^*)$ through $\tau^*(t^*)$. Thus we must solve (12) and (13) in order to solve (22)–(24). Note that Z^*, which indicates distance from the symmetry axis, does not appear in (22)–(24), either implicitly or explicitly. Within our approximations, which are valid only for $|Z^*| \ll 1$, depletion layer structure is independent of Z^*.

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NUMERICAL SOLUTIONS FOR FLOW ALONG THE MAGNETIC FIELD

Characteristics. We solve (12) and (13) using the method of characteristics [*Werner*, 1968]. This requires the initial values $a^*(y^*, 0)$ and $v^*(y^*, 0)$ and the characteristics λ_c of the differential equations. The latter can be obtained by noting that (12) and (13) can be combined and written in tensor notation as

$$\delta_{ji}\, \partial/\partial t^*\, u^i + A_{ji}\, \partial/\partial y^*\, u^i = G_j \tag{25}$$

where **u** is a column vector whose elements are $u^1 = a^*$ and $u^2 = v^*$. Indices i and j can take the values 1 or 2, and repeated indices are summed over. Applying the undetermined multipliers $\lambda^{\nu j}$ to (25) yields

$$\lambda^{\nu j}\delta_{ji}(\partial u^i/\partial t^* + \lambda_c\, \partial u^i/\partial y^*) = \lambda^{\nu j}G_j$$

where λ_c is defined by

$$\lambda^{\nu j}(\delta_{ji}\lambda_c - A_{ji}) = 0 \tag{26}$$

Equation (26) is homogeneous in the multipliers $\lambda^{\nu j}$ and has a nontrivial solution only if the determinant of the coefficients vanishes. Since (12) and (13) imply that $A_{11} = A_{22} = v^*$, the nontrivial solution of (26) is

$$\lambda_c = v^* \pm (A_{12}A_{21})^{1/2} \tag{27}$$

It can be shown [*Zwan*, 1974] that

$$(A_{12}A_{21})^{1/2} = C_{A0}^{-1}[C_A^2C_S^2/(C_A^2 + C_S^2)]^{1/2} \tag{28}$$

where C_A and C_S are the Alfvén speed and the sound speed, respectively. We point out that (28) is identical with the apparent phase velocity along the magnetic field of a slow mode wave traveling almost perpendicular to the ambient magnetic field. The disturbance (i.e., the reduction in particle pressure and corresponding increase in magnetic pressure due to squeezing) propagates as a slow mode wave.

Initial value arrays. Along with the characteristics given by (27) we need the initial values $a^*(y^*, 0)$ and $v^*(y^*, 0)$ in order to solve (25). We start the model flux tube such that ρ_0, the density at $y = 0$ and $t = 0$, is related to ρ_{SW}, the preshock solar wind density, by

$$\rho_0 = 4\rho_{SW} \tag{29}$$

as is expected for a strong shock with $\gamma = \frac{5}{3}$.

To find $a^*(y^*, 0)$, we combine (3) and (4) and write the external pressure in the form

$$p_{ext}(y) = K\rho_0^{\gamma}\left(\frac{a_0}{a(y, 0)}\right)^{\gamma} + p_{M0}\left(\frac{a_0}{a(y, 0)}\right)^2 \tag{30}$$

where, as usual, the subscript 0 denotes a quantity evaluated at $y = 0$ and $t = 0$ and $a(y, 0)$ is the initial value of a as a function of y. With (3*b*), (5), and (9), (30) can be written

$$p_{ext}^*(y^*) = (1 - \xi)\left(\frac{a_0^*}{a^*(y, 0)}\right)^{\gamma} + \xi\left(\frac{a_0^*}{a^*(y, 0)}\right)^2 \tag{31}$$

Considering that (5) and (9) imply $a_0^* = \xi^{-1/2}$ and using (11) and $\gamma = \frac{5}{3}$, we can write (31) as

$$[a'(y^*)]^6 + (1 - \xi)\xi^{-5/6}[a'(y^*)]^5 - (1 + y^{*2})^{-1} = 0 \tag{32}$$

given the definition $a'(y^*) \equiv [a^*(y^*, 0)]^{-1/3}$. Equation (32) is solved numerically at every point y^*, taking the largest positive real root for the initial value array $a^*(y^*, 0)$.

For the initial velocity array we use a somewhat complicated formula, which in the important region with $|y^*| \ll 1$ reduces to

$$v^*(y^*, 0) \approx 1.14y^*\xi^{-1/2} \tag{33}$$

(see Appendix D). Most of this initial velocity comes from the fact that as the tube crosses the shock, the plasma receives a small 'kick' out along the field lines because **B** is tangent to the bow shock only at $y^* = 0$. A second effect that has been included in (33) is a correction for noninertiality of the *xyz* coordinate system.

Solutions for flow along the magnetic field. We now have the wherewithal to compute motions along a magnetic field line by solving the partial differential equations (12) and (13) with the characteristics given by (27) and the initial value arrays obtained from (32) and (33). To determine the solution, we need only specify ξ, the ratio of magnetic pressure, just postshock at $y = 0$ to total external pressure; the parameter can be related to the solar wind Alfvén Mach number by using (9), (B2), and (D3):

$$M_A(SW) = 3.01\xi^{-1/2} \tag{34}$$

Figures 4 and 5 show the solutions of (12) and (13) for $M_A(SW) = 8.0$, i.e., $\xi = 0.141$. Figure 4 is a plot of the

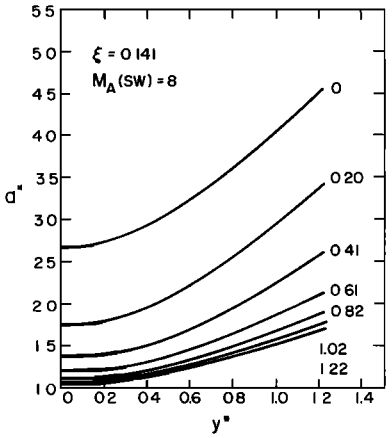

Fig. 4. Solutions of (12) and (13) for the dimensionless cross-sectional area a^* for the case of $M_A(SW) = 8$. The number at the end of each curve represents the t^* value for that particular curve.

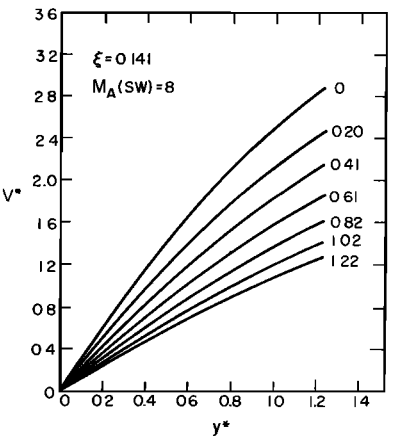

Fig. 5. Solutions of (12) and (13) for the dimensionless plasma velocity V^* in the direction of the field for the case of $M_A(SW) = 8$. The number at the end of each curve represents the t^* value for that particular curve.

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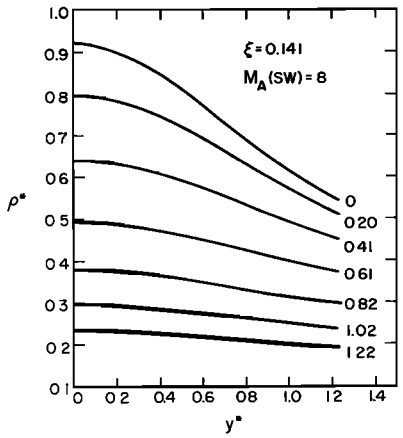

Fig. 6. The dimensionless plasma density obtained from the solutions of (12) and (13) and (7) and (11) for the case of $M_A(SW) = 8$. The number at the end of each curve represents the t^* value for that particular curve.

dimensionless flux tube cross-sectional area a^* versus the dimensionless distance along the field line y^*. The quantities can be transformed to a and y with (5). The different curves in Figure 4 represent solutions of (12) and (13) at different t^* values. The corresponding values of the dimensionless velocity v^* are plotted in Figure 5. Combining (11) and (7), we calculate the ρ^* solutions which are given in Figure 6.

Figures 7 and 8 show plots of two quantities that are directly needed in the calculation of $D^*(t^*)$ from (22): $a^*(0, t^*)/a_0^*$, the ratio of the initial magnetic field to the magnetic field at t^*, both for $y^* = 0$ and for the depletion factor

$$d_f(t^*) = \rho_0^*/\rho^*(0, t^*) = \rho_0/\rho(0, t^*) \tag{35}$$

the ratio of the initial density to the density at t^*, both at $y^* = 0$.

Depletion Layer Thickness—Numerical Results

In this section we evaluate the integral in (22) numerically as a function of t^* to determine $D^*(t^*)$, the dimensionless distance from the magnetopause. Comparing this result with Figure 8 allows us to plot the depletion factor d_f versus the

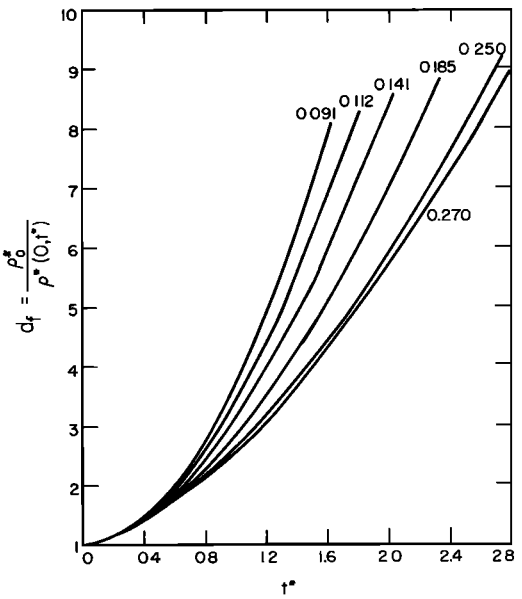

Fig. 8. A plot of the depletion factor $d_f(t^*) = \rho(0, 0)/\rho(0, t^*)$, the ratio of the initial density to the density at dimensionless time t^*, at the center of the flux tube ($y^* = 0$).

dimensionless distance from the magnetopause, the basic result of the paper (see Figures 9 and 10).

Evaluation of factors in (22). The factor $\tau^*(t^*)$ is computed from

$$\tau^*(t^*) \equiv C_{A0}\tau(t^*)/l = [\xi\rho(0, t^*)/\rho_0]^{1/2}\sigma^{-1} \tag{36}$$

which follows from (19), (16), and (11) by using definitions from (5) and (9). The integral I is computed numerically by using a seven-term formula,

$$\rho^*(0, t^*) = \sum_{m=1}^{N} C_m(\xi)t^{*m-1} \tag{37}$$

to fit the densities computed, e.g., see Figure 8. A similar seven-term formula is used to fit the ratio $a^*(0, t^{*\prime\prime})/a_0^*$ computed for Figure 7. To evaluate S^*, we set v_{X0} equal to $\frac{1}{4}$ the

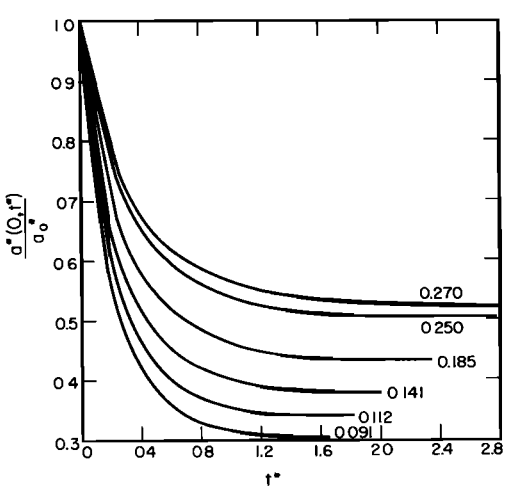

Fig. 7. The ratio of the dimensionless flux tube cross-sectional area at the origin at time t^* to the same dimensionless quantity at $t^* = 0$ versus t^*. The number at the end of each curve is the value of ξ for that particular curve.

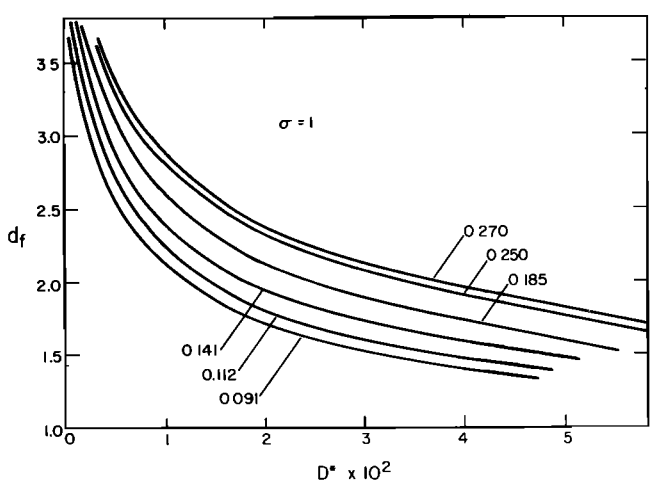

Fig. 9. The depletion factor versus the dimensionless distance from the magnetopause for the calculation excluding tension ($\sigma = 1$). The number on each curve represents the ξ value for that particular curve.

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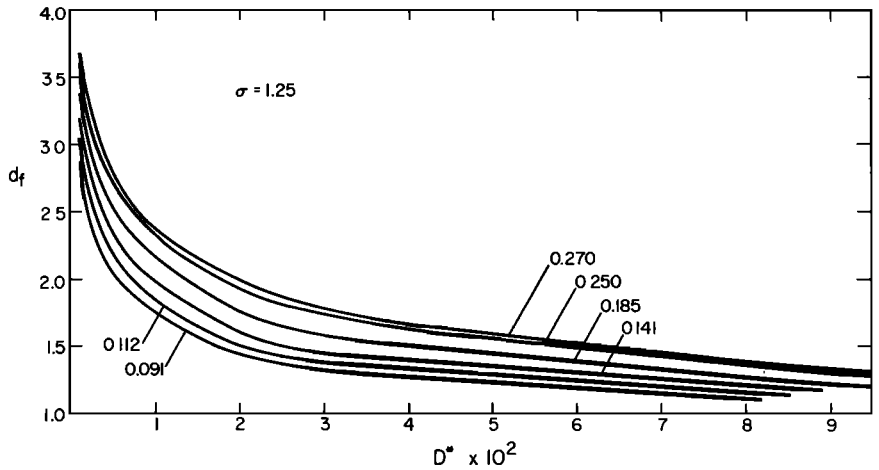

Fig. 10. The depletion factor versus the dimensionless distance from the magnetopause for the calculation including tension (σ = 1.25). The number on each curve represents the ξ value for that particular curve.

solar wind speed (see (D4)); the ratio v_{z0}/Z_0 is derived from an analysis of the kick that the bow shock gives the plasma in the Z direction. S^* is always near unity, and its detailed evaluation for different ξ values is discussed in Appendix D. One final comment must be made concerning the numerical integration over $t^{*\prime\prime}$ in (22): for obvious practical reasons the infinite upper limit was replaced by a finite one, namely, $t_{max}^* \approx 5.4\tau_0^*$. We have estimated the resulting error and find it to be negligible for depletion factors $d_f \lesssim 4$.

Computed density profiles. Figures 9 and 10 show the results of evaluating $D^*(t^*)$ from (22). Plotted on the ordinate is the depletion factor d_f rather than t^*, the relation between d_f and t^* having been plotted in Figure 8. The curves shown in Figure 10 are computed for σ = 1.25, which we feel represents an overestimate of the tension effect (see Appendix C). To convert D^* to kilometers, multiply by the scale length l, which is approximately 72,500 km for a magnetosphere with a magnetopause standoff distance of 10 R_E (see Appendix B). The depletion factor d_f is defined in (35). The perspective of Figures 9 and 10 is such that the bow shock would be to the far right (off the scale), and the magnetopause could be considered the ordinate axis, i.e., D^* = 0. Note that the depletion layer gets thicker as ξ increases, a physically reasonable result. Small ξ implies an initial magnetic energy much smaller than the particle energy, so that a very large fraction of the compressed plasma must be squeezed out of the center of the flux tube before particle and magnetic pressures can become comparable; the larger the fraction of plasma that must be squeezed out along field lines, the longer the required squeezing time and the nearer the flux tube must come to the stagnation point and the magnetopause.

It is apparent that the inclusion of magnetic tension in the calculation might reduce the depletion layer thickness by as much as a factor of 2. Physically, tension should cause the flux tube to accelerate parallel to the magnetopause boundary away from the subsolar point. For a given residence or convection time this acceleration will result in the flux tube residing closer to the magnetopause, i.e., D^* will be smaller. Actual values of D^* should lie between the extreme values computed for σ = 1.25 and σ = 1.00.

Table 1 gives other results of the same series of calculations. Column 2 gives Alfvén Mach numbers in the solar wind computed from (34). Column 6 lists $t_{1/2}^*$, the dimensionless time required for the density at y^* = 0 to drop by a factor of 2 below its initial postshock value. Dimensionless and dimensional depletion layer thicknesses are listed in the last four columns for σ = 1.00 and σ = 1.25; the degree of depletion d_f is a continuously decreasing function of distance from the magnetopause, but we define the depletion layer thickness as the thickness of the layer whose depletion factor exceeds 2. In Table 1 the dimensional quantities C_{A0}, $t_{1/2}$, and D are evaluated for l = 72,500 km, $p_0 = 1.18 \times 10^{-8}$ dyn cm^{-2}, and $\rho_0 = 3.3 \times 10^{-23}$ g cm^{-3}. Dimensionless quantities depend only on ξ. Figure 11 is a plot of depletion layer thickness versus ξ. Note that the thickness is approximately proportional to ξ.

SIMPLE FORMULA FOR DEPLETION LAYER THICKNESS

We derive a nonrigorous approximate formula for the depletion layer thickness, specifically the thickness of the layer in which the density has been reduced by a factor of 2 or more. If the density drops a factor of 2, so that the gas pressure p_g drops a factor of $2^{5/3}$, the cross-sectional area a of the flux tube

TABLE 1. Results of Calculations of Depletion Layer Thickness

							D^*		D, km	
ξ	$M_A(SW)$	C_{A0}, km s^{-1}	τ_0^*	t_{max}^*	$t_{1/2}^*$	$t_{1/2}$, min	σ = 1.00	σ = 1.25	σ = 1.00	σ = 1.25
0.091	10.0	80.1	0.302	1.64	0.60	9.05	0.013	0.006	900	500
0.112	9.0	88.9	0.335	1.82	0.63	8.57	0.015	0.008	1100	600
0.141	8.0	99.7	0.376	2.04	0.66	7.99	0.019	0.010	1300	700
0.185	7.0	114.2	0.430	2.34	0.70	7.41	0.024	0.013	1800	1000
0.250	6.0	132.8	0.500	2.71	0.73	6.64	0.033	0.019	2400	1400
0.270	5.8	138.0	0.520	2.82	0.74	6.48	0.036	0.021	2600	1500

The d_f is 2 for dimensionless (D^*) and dimensional (D) depletion layer thicknesses.

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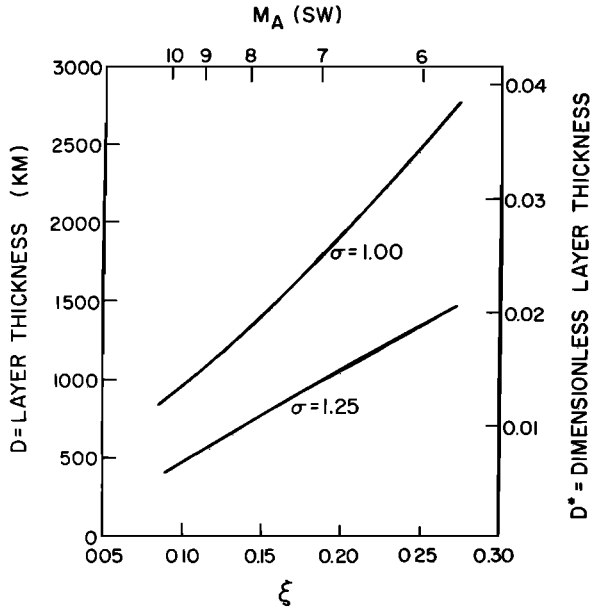

Fig. 11. The depletion layer thickness versus ξ, the ratio of magnetic pressure just postshock to total pressure at the magnetopause. D and D^* give the thickness of the layer where the density has been reduced by a factor of 2 or more. The thicknesses D in kilometers apply to a magnetopause standoff distance of 10 R_E. The quantity $M_A(SW)$ is the ratio of flow speed to Alfvén speed in the preshock solar wind. The curves marked $\sigma = 1$ and $\sigma = 1.25$ represent results of extreme opposite assumptions about effects of magnetic tension.

has to decrease enough to allow the magnetic pressure to get close to the total external pressure p_0 ($\simeq \rho_{SW} v_{SW}^2$).

$$B^2/8\pi = B_0^2/8\pi(a_0/a)^2 \simeq p_0 \tag{38}$$

where, as usual, $\xi = B_0^2/(8\pi p_0)$ (see (9)). Thus we have

$$a/a_0 \simeq \xi^{1/2} \tag{39}$$

where, as was the case before, the subscript zero on B and a refers to conditions at $t = 0$ just postshock.

To estimate the time required for sufficient reduction in a, we note that the equation of motion of a fluid element in the y direction near the external pressure maximum at $y = 0$ can be written

$$d^2y/dt^2 \approx -y(\rho)^{-1}\ \partial^2 p_g/\partial y^2 \equiv y\tau_y(t)^{-2} \tag{40}$$

which has exponentially growing solutions proportional roughly to $\exp\left[\int_0^t dt'/\tau_y(t')\right]$. Thus v_y at given y is given by

$$v_y \simeq y/\tau_y \tag{41}$$

We can estimate the time required for a to drop by the amount required in (39) by applying conservation of mass between $y = 0$ and some small y value:

$$\begin{aligned} y\ d(\rho a)/dt &\cong -(\rho a y/\tau_y) \\ \rho a/\rho_0 a_0 &= \exp\left[-\int_0^t dt'/\tau_y(t')\right] \end{aligned} \tag{42}$$

We set $\rho/\rho_0 = \frac{1}{2}$ and eliminate a/a_0 from (39) and (42), obtaining

$$\exp\left[-\int_0^{t_{1/2}} dt'/\tau_y(t')\right] \cong \tfrac{1}{2}\xi^{1/2} \tag{43}$$

which could be solved for the time $t_{1/2}$ corresponding to a factor of 2 depletion. The time $t_{1/2}$ decreases with increasing ξ, since as ξ increases, a lesser reduction in flux tube area is required for the magnetic pressure to become comparable to the particle pressure.

A flux tube at the magnetopause has resided in the magnetosheath an infinite time (ideally); this residence time t decreases as D, the distance from the magnetopause, increases. We found earlier that the derivative of D with respect to t was given by

$$|(\partial D/\partial t)_Z| = (|v_{X0}|B_0/v_Z B)|(\partial Z_0/\partial t)_Z| \tag{44}$$

where v_{X0} and B_0 are the fluid velocity and the magnetic field just postshock and v_Z and B are the velocity (along the magnetopause) and the magnetic field in the depletion layer at distance Z from the stagnation streamline (cf. (15)). Brushing past subtleties like initial velocities and tension effect, we find that the equation of motion in the Z direction,

$$d^2Z/dt^2 = -1/\rho(\partial^2 p_{ext}/\partial Z^2)Z \equiv Z/\tau_Z(t)^2 \tag{45}$$

has the approximate solution

$$Z = Z_0 \exp\left[\int_0^t dt'/\tau_Z(t')\right] \tag{46}$$

thus

$$\left|\frac{1}{v_Z}\left(\frac{\partial Z_0}{\partial t}\right)_Z\right| \sim \exp\left[-\int_0^t dt'/\tau_Z(t')\right] \tag{47}$$

The thickness of a layer with residence time greater than t is given by

$$D(t) \sim \int_t^\infty dt' \frac{|v_{X0}|\ B_0}{B(t')} \exp\left[-\int_0^{t'} dt''/\tau_Z(t'')\right]$$

Replacing $\tau Z(t'')$ by a constant $\bar{\tau}_Z$ and getting $B(t)$ from (38) and (39) yield

$$D(t) \sim \bar{\tau}_Z|v_{X0}|\xi^{1/2} \exp(-t/\bar{\tau}_Z) \tag{48}$$

Replacing $\int_0^{t^{1/2}} dt'/\tau_y(t')$ by $t/\bar{\tau}_y$ in (43) and substituting the result in (48) yield for the thickness of the depletion layer,

$$D \sim 2\bar{\tau}_Z|v_{X0}|(\xi/4)^g \tag{49}$$

where

$$g = (1 + \bar{\tau}_y/\bar{\tau}_Z)/2$$

Using $p_0 \simeq \rho_{SW} v_{SW}^2$, the strong shock relations (29), (D3), and (D4), the relation $d^2p_{ext}/dZ^2 \simeq -2p_0/l^2$, and $\bar{\tau}_Z \simeq \tau_{Z0}$, we obtain

$$D \sim (l/2^{1/2})(2/M_A^2)^g \tag{50}$$

where M_A is the Alfvén Mach number in the solar wind. The times $\tau_y(t)$ and $\tau_Z(t)$ are approximately equal for small t; it is not easy to estimate their ratio for larger t, much less the complicated weighted amount $\bar{\tau}_y/\bar{\tau}_Z$. Simply setting $\bar{\tau}_y/\bar{\tau}_Z = 1$ gives

$$D \sim (2^{1/2})l/M_A^2$$

The same formula, but with a slightly different coefficient, gives a reasonable fit to the $\sigma = 1$ curve in Figure 11:

$$D \simeq 1.24l/M_A^2 \tag{51}$$

where D is the thickness of the layer in which the density has been reduced by a factor that is ≥ 2. The quantity l, measured along the magnetopause, is the distance from the symmetry axis to the point where the total pressure at the magnetopause

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(p_{ext}) is half as large as that at the symmetry axis; for the earth's magnetosphere and a magnetopause standoff distance of 10 R_E we have l = 72,500 km. The quantity M_A^2 is $v_{SW}^2 4\pi\rho_{SW}/B_{SW}^2$, the square of the Alfvén Mach number in the preshock solar wind.

DISCUSSION AND SUMMARY

Comparison with Lees' model. *Lees* [1964] computed the depletion layer thickness on the stagnation streamline for one case, and we can compare our results directly with his. Although our (22), which relates depletion time with distance from the magnetopause, was derived for location Z, substantially off the stagnation streamline, it can be shown [*Zwan*, 1974] that the same equation holds within the same approximations on the stagnation streamline. In Figure 12 we have compared Lees' results with ours for $M_A(SW)$ = 5.8, i.e., ξ = 0.27. Lees' data are normalized to a bow shock standoff distance of approximately 23,000 km. His model differs from ours in that he assumed $\gamma = 2$ instead of $\frac{5}{3}$, but more importantly, it differs in that he assumed completely axisymmetric flow. Close to the boundary the axisymmetric velocity approximation would seem to overestimate the velocity of plasma out along field lines, while it underestimates the velocity of the flux tube parallel to the boundary. This combination would yield thicker depletion layers and explain why Lees' values of ρ/ρ_2 are smaller than our depletion model values for small D (see Figure 12). In our model the departure from axisymmetry is due to the fact that the complete $\partial p_{ext}/\partial Z$ is available to accelerate plasma in the Z direction, whereas only the smaller $\partial p_{gas}/\partial y$ is available to accelerate plasma parallel to y, and the latter becomes small in the depletion layer when $p_{gas} \lesssim p_{mag}$.

One might think that our assumption of axisymmetric p_{ext} and Lees' assumption of totally axisymmetric flow in the depletion layer would possess equivalent levels of inaccuracy, but that is not the case. Let us neglect the small departures of the shape of the magnetosphere from axisymmetry due to the orientation of the earth's dipole and consider just deviations from axisymmetry caused by the orientation of the interplanetary magnetic field. Total pressure p_{ext} in the magnetosheath should be axisymmetric except for corrections of the order of ξ; that is, deviations should be ~10% for $M_A(SW)$ ~10 and would go to zero as $M_A(SW)^2 \to \infty$. Flow velocities outside the layer would deviate from axisymmetry to the same small degree. Force balance normal to the thin depletion layer requires that $p_{ext} = p_g + B^2/8\pi$ be nearly constant through the thickness of the layer for given y and Z; thus if p_{ext} is axisymmetric outside the layer, it is axisymmetric inside. The same is not true for v_y, which can vary through the layer, which is a finite thickness tangential discontinuity. Flow velocities in the layer deviate substantially from axisymmetry. For example, for a depletion factor of 2 we find $v_Z^*(y^* = 0, Z = r^*) \approx 2v_y^*(y^* = r^*, Z = 0)$ for $r^* \lesssim 0.5$ and $5.8 \leq M_A(SW) \leq 10$. The ratio of these two velocities does not approach unity as $M_A(SW) \to \infty$.

Comments on the computed layer thicknesses. The layer thickness is a smooth function of y and z near the subsolar point, and, in fact, $\partial D/\partial y = dD/dZ = 0$ at that point. Within the approximations made here, which are valid only for $|y|$ and $|Z|$ much less than $l \approx 11\ R_E$, the layer thickness is uniform on the day side magnetopause. The region where $B^2/8\pi \gtrsim p_{gas}$ is basically a thin layer covering the whole magnetopause not a hemispherical region with radius ~0.5 R_E, as was suggested by *Alksne* [1967].

The layer thickness increases as the solar wind Alfvén Mach number decreases and increases approximately linearly with increasing ξ; see Figure 11.

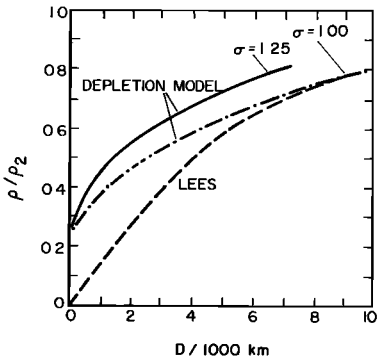

Fig. 12. A comparison of our density profiles with one computed by *Lees* [1964]. All curves are for $M_A(SW)$ = 5.8; we used $\gamma = \frac{5}{3}$, but Lees used γ = 2. The density ρ_2 is the density just postshock, so that $\rho/\rho_2 = d_f^{-1}$.

Competition between depletion and magnetic merging. Extremely low densities cannot exist in the depletion layer in the presence of magnetic merging because magnetosheath flux tubes that come close to the magnetopause would be merged to geomagnetic field lines before they had a chance to be depleted fully. For $\rho^*/\rho_0^* \gtrsim 0.5$ the relationship between density and residence time (plotted in Figure 8) can be written approximately as

$$\rho^*/\rho_0^* \approx 1 - 0.77t^* \tag{52}$$

A time $t^* \approx 0.66$ corresponds to a factor of 2 depletion for M_A = 8.

To estimate magnetosheath residence time in an open magnetosphere model, consider a flux tube whose equatorial crossing point moves along the symmetry line (X axis) from the bow shock ($X \approx 1.3D_m$) to the point m just sunward of the merging region ($X \equiv 1.0D_m$) (see Figure 1). Assume that v_X is approximately linear in X, and let v_m be the 'merging speed'; the velocity at $X = D_m$ is

$$|v_X| = v_m + (|v_{X0}| - v_m)(X - D_m)/0.3D_m$$

The time required to go from $X = 1.3D_m$ to $X = 1.0D_m$ is then given by

$$t_m^* = \frac{C_{A0}}{l}\int_{D_m}^{1.3D_m}\frac{dX}{|v_X|} = \frac{0.3D_mC_{A0}}{(|v_{X0}| - v_m)l}\ln\left(\frac{|v_{X0}|}{v_m}\right)$$

Putting $D_m \approx 0.88l$ (which follows from Spreiter's flow patterns and pressure diagrams) and

$$|v_{X0}| \approx 0.38\xi^{-1/2}C_{A0} \tag{53}$$

(which follows from (29), (B2), and (D4)), we obtain

$$t_m^* = 0.7\xi^{1/2}(1 - \zeta)^{-1}\ln(\zeta^{-1}) \tag{54}$$

where $\zeta = v_m/|v_{X0}|$. Let $v_m = kC_{Am}$, where C_{Am} is the Alfvén speed just outside the merging region and k is a constant ~0.1 or ~0.2 [*Vasyliunas*, 1974, 1975; *Sonnerup*, 1974; *Hill*, 1975]. Using (52) and (3*b*), we find that

$$\left(\frac{C_{Am}}{C_{A0}}\right)^2 = \frac{1 - (1 - \xi)(1 - 0.77t_m^*)^{5/3}}{\xi(1 - 0.77t_m^*)} \tag{55}$$

and using (53),

$$\zeta = 2.655\xi^{1/2}kC_{Am}/C_{A0} \tag{56}$$

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TABLE 2. Interaction Between Depletion and Magnetic Merging

ξ	k	t_m^*	ρ_m/ρ_0	C_{Am}/C_{A0}
0.141	0.05	0.59	0.55	3.0
0.141	0.10	0.48	0.63	2.6
0.141	0.20	0.38	0.71	2.3
0.091	0.10	0.40	0.69	2.9
0.270	0.10	0.61	0.53	2.3

Table 2 shows the results of substituting (56) and (55) into (54) and solving numerically for t_m^* for various values of ξ and k. Given t_m^* it is, of course, easy to calculate ρ_m/ρ_0, which represents the degree of depletion just outside the merging region, and C_{Am}/C_{A0}, which indicates the degree to which depletion has increased the Alfvén speed. Three conclusions are obvious: (1) depletion layers with depletion factors $d_f \geq 2$ are unlikely to form unless the merging coefficient k is small, (2) for $k \sim 0.1$ the squeezing process can reduce densities ~30%, and (3) depletion can substantially increase the Alfvén speed and thus possibly the merging rate, as *Vasyliunas* [1975] pointed out to us.

Relationship to observations. In a magnetometer record a depletion layer is indicated if the magnitude of the field is approximately the same on the sunward and earthward sides of the magnetopause as defined by an abrupt field angle change. If the magnetopause is approximated by a tangential discontinuity and if the field magnitude is constant across the discontinuity, the particle pressure on both sides should be approximately the same thus yielding an anomalously high field strength in the magnetosheath adjacent to the magnetopause. We define 'anomalously low or high' with respect to what one would normally expect for compressed solar wind. A rough survey of published magnetometer data from magnetopause crossings [*Zwan,* 1973] suggests that the depletion layer signature occurs quite commonly on the order of half the crossings.

Of course, a depletion layer can be identified reliably in a single satellite crossing only if simultaneous plasma and magnetic field data are available. There are relatively few published records of such simultaneous data during a magnetopause crossing. One case that has been exhaustively studied is the ATS 1 magnetopause encounter of January 13–14, 1967; in that case, *Cummings and Coleman* [1968] saw a clear depletion layer signature in the magnetic field, and *Freeman et al.* [1968] observed an anomalously low plasma density in the magnetosheath just outside the magnetopause.

Depletion layers near other planets. A depletion layer should tend to form in front of any 'stationary' obstacle in the solar wind if the obstacle is effectively impenetrable. Our calculations are easily scaled to any approximately spherical obstacle: for given ξ value (or solar wind Alfvén Mach number) the depletion layer thickness is proportional to l, the distance along the boundary to the half-pressure point, which is a single measure of the size of the obstacle. Such scaling to Mars and Venus suggests layer thicknesses about $\frac{1}{15}$ and $\frac{1}{10}$ that of the earth, respectively. However, diffusion may obliterate the Martian and Cytherean depletion layers.

APPENDIX A: EFFECT OF CROSS-FIELD DIFFUSION ON A DEPLETION LAYER

To evaluate the probable effect of cross-field diffusion on the depletion layer, we compare the diffusive flux of particles into a depletion layer with the flux of particles squeezed out of the layer along field lines.

Consider specifically a section of the depletion layer with $|y| < l_y/2$ and $|Z| < l_z/2$. The number of particles diffusing into the layer per unit time can be written $l_y l_z D|\nabla n|$, where D is the diffusion coefficient and $|\nabla n|$ is the density gradient just outside the layer. Suppose that D equals the Bohm coefficient $ckT/16eB$, with $B = 50\,\gamma$ and $T = 4 \times 10^6$°K so that $D = 4 \times 10^{12}$ cm^2 s^{-1}. Taking $|\nabla n| \sim 6.7 \times 10^{-8}$ cm^{-4} from our depletion calculations and $l_y = l_z = 10\,R_E$, we obtain for the diffusive flux of particles into the layer $\sim 1 \times 10^{25}$ s^{-1}.

The number of particles squeezed out of the ends of the flux tubes of the layer per second is given by $2l_z n|v_y|d_c$, where n and v_y are the number density and the parallel velocity at the ends of the flux tubes and d_c is the layer thickness. If we take $n \sim 10$ cm^{-3}, $|v_y| \sim 100$ km/s, and $d_c \sim 1500$ km, the number of particles squeezed out per unit time is $\sim 2 \times 10^{26}$ s^{-1}.

Considering that the assumption of Bohm diffusion probably overestimates the diffusion flux we conclude that cross-field diffusion is unlikely to affect the depletion layer importantly.

APPENDIX B: ASSUMPTIONS ABOUT EXTERNAL PRESSURE

For our treatment of motion in the y direction we set

$$p_{ext}^* = p_{ext}/p_0 = (1 + y^{*2})^{-1} \tag{B1}$$

where $y^* = y/l$. To match the results of *Spreiter et al.* [1966], we set

$$p_0 = 0.881\rho_{SW}v_{SW}^2 \tag{B2}$$

For an average magnetosphere with a magnetopause standoff distance of 10 R_E we have $p_0 \approx 1.18 \times 10^{-8}$ dyn cm^{-2}, and $l \approx 7.25 \times 10^9$ cm. Equation (B1) is assumed to be independent of time. In our treatment of motion in the Z direction we assume $\partial^2 p_{ext}^*/\partial Z^{*2} = \partial^2 p_{ext}^*/\partial y^{*2} = -2$ at $y^* = Z^* = 0$.

Several aspects of this assumed external pressure distribution require discussion.

1. Our computed depletion layer thicknesses, which apply to $y^* = 0$, are completely insensitive to the values of p_{ext}^* chosen for $y^* \gtrsim 1$ because the flow is effectively supersonic for larger y^*. Information propagates through the flowing gas at the slow mode speed given in dimensionless form by (28); this speed is usually ~1 in dimensionless units for the cases treated. For example, consider the case $\xi = 0.141$ illustrated in Figures 4–6. For $t^* < 1.22$, the only times considered, one can easily show by integrating wave motions over time that the origin ($y^* = 0$) cannot receive messages from $y^* > 0.7$. Times $t^* > 1.22$ occur only in layers too thin to be physically significant. Thus the computed depletion layer thicknesses could not change if we chose different values for p^* at larger y^* except possibly if we took a physically unreasonable model in which p^* increased dramatically with y^* at large y^*, causing a shock that might be able to propagate against the stream. The features of the external pressure distribution that have important physical effects on flow within a flux tube are the external pressure and its second derivative at $y^* = 0$, parameters that enter our formalism just as scale factors connecting dimensionless and dimensional quantities; our numerical results can trivially be scaled to a wide range of values of p_0 and l.

2. The assumption that p^* is independent of t^* neglects the fact that p^* would actually increase slightly near $y^* = Z^* = 0$ as the tube moved from the shock to the depletion layer. From the flow patterns of *Spreiter et al.* [1966] we estimate that the error in p^* that results from putting p^* equal to its magnetopause value beginning at the time of shock crossing ($t^* = 0$) is

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$\approx 16\%$ at $t = t^* = 0$ but drops rapidly, reaching 5% by $t \approx 80$ s ($t^* \sim 0.1$).

3. For approximate axial symmetry, p^* would be represented better by $(1 + y^{*2} + Z^{*2})^{-1}$ than by (B1). Since we neglected Z^{*2} in computing flow within the tube, our results are only accurate for $Z^{*2} \lesssim 0.1$ or for $Z \lesssim 4\ R_E$.

4. We suggest using pressures computed by *Spreiter et al.* [1966] to determine the scale factors p_0 and l because we feel that their gasdynamic calculations are reasonably good approximations to the magnetosheath flow outside the depletion layer; the magnetic field usually does not contribute importantly to the total stress tensor there, and the gasdynamic calculations fit the observations very well [*Spreiter and Alksne*, 1969]. The use of gasdynamic values of p_{ext} in calculations of depletion layer structure, of course, approaches exact validity as $M_A(SW) \to \infty$.

5. One source of error in the use of the gasdynamic model information to specify p_{ext} lies in the approximation of axial symmetry, specifically $\partial^2 p_{ext}/\partial y^2 = \partial^2 p_{ext}/\partial Z^2$ at $y = Z = 0$. We thus follow most authors in approximating the magnetosphere as a body of revolution. Departure of p_{ext} from axisymmetry owing to effects of the magnetosheath magnetic field is hard to estimate but is presumably small, since the ratio of magnetic stress to particle stress is usually small throughout the day side magnetosheath (except for the depletion layer). Axisymmetry of p_{ext} becomes exact as $M_A(SW) \to \infty$. We feel that the uncertainty in Z acceleration owing to the unknown magnetic tension effect (see (19) and Appendix C), a major uncertainty that is explicitly acknowledged in our quoted layer thicknesses, is likely for $M_A(SW) \gtrsim 6$ to be larger than the error we make in assuming p_{ext} axisymmetric.

APPENDIX C: MAGNETIC TENSION ANALYSIS

Inclusion of magnetic tension in the analysis should decrease the layer thickness. Tension in the magnetic field lines is expected to accelerate flux tubes essentially parallel to the magnetopause boundary away from the X axis and the subsolar point. Streamlines that intersect the bow shock close to the equatorial plane converge toward the magnetopause. Thus the additional acceleration imposed by the inclusion of tension results in a flux tube for a given convection time being convected farther along a given streamline and therefore closer to the magnetopause.

Although precise inclusion of magnetic tension into the layer thickness analysis is difficult, we can estimate limits on its effects. We bracket our results by considering the following cases: (1) no tension and (2) maximum effect of tension. The effect of tension depends on how the field lines drape themselves around the magnetosphere. We assume that the actual geometry is between the two extremes labeled (*a*) and (*b*) in Figure 13. The figure is a projection onto the X-Z plane. The magnetosphere near the subsolar point is assumed to have a radius of curvature R; the center of the equivalent sphere of radius R lies behind the earth. In configuration (a), a depletion layer field line is assumed to approximate an arc of a great circle on the surface of the sphere; the curvature vector of the field line is perpendicular to the magnetopause, so that the magnetic tension force has no component parallel to the boundary. However, in configuration (b), depletion layer field lines lie in a plane parallel to the earth-sun line, and there is a component parallel to the boundary. The tension force per unit volume F_t is $B^2/4\pi R$, and the Z component, which is defined to be the component parallel to the boundary, is given by

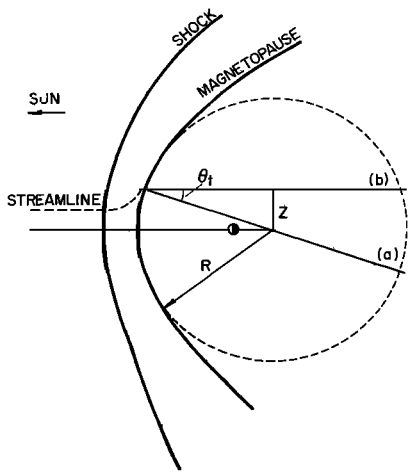

Fig. 13. A sketch of magnetic field lines in our two extreme assumptions about the effects of tension. In configuration (*a*), tension plays no role in the acceleration of the flux tube along the boundary; configuration (*b*) leads to a maximum tension effect. The figure is a projection onto the X-Z plane. The magnetosphere has radius of curvature R.

$$\mathbf{F}_t \cdot \hat{Z} = (B^2/4\pi R) \sin \theta_t \tag{C1}$$

From Figure 13, $\sin \theta_t = Z/R$, and we can write the tension term as

$$\mathbf{F}_f \cdot \hat{Z} \approx (B^2/4\pi R^2) Z \tag{C2}$$

We can place an upper limit on the tension effect by choosing

$$B^2/4\pi R^2 = 2p_M/R^2 = \lim_{p_{ext} \to 0} 2p_0/R^2 \tag{C3}$$

By using (11) and (C3), (19) can be written

$$\tau(t) = [\rho(t)/2p_0]^{1/2}[R^2 l^2/(R^2 + l^2)]^{1/2} \tag{C4}$$

where R is the radius of curvature of the subsolar magnetopause. For an average magnetopause with $l = 11.36\ R_E$ (see appendix B) we have $R \simeq 15\ R_E$, so that (C4) becomes

$$\tau(t) = (l/\sigma)[\rho(t)/2p_0]^{1/2} \tag{C5}$$

where $\sigma = 1.25$ for calculation of the upper limit of the tension effect; to neglect and therefore to underestimate the tension effect, we put $\sigma = 1.00$.

APPENDIX D: INITIAL CONDITIONS

Initial conditions on v_y. We use our ability to specify the initial values of v_y as functions of y to include in the calcu-

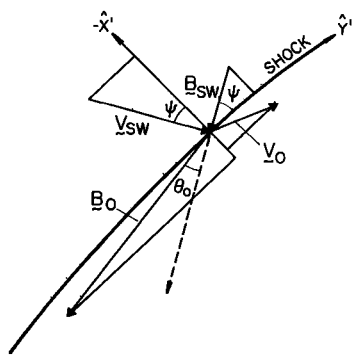

Fig. 14. The coordinate system and the parameter orientation used in the calculation of the initial field aligned velocity due to bow shock obliquity.

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lations the initial kick that the bow shock gives plasma parallel to the magnetic field and the effects of the noninertiality of the xyz coordinate system, which $\mathbf{E} \times \mathbf{B}$ drifts with the magnetic flux tube:

$$v_y = v_{\text{kick}}(y, 0) + v_{nt}(y, 0)$$

Initial velocity owing to bow shock obliquity. To discuss jump conditions across the shock, it is convenient to define an $X'Y'Z'$ coordinate system locally at a point on the shock. The system is locally orthogonal, the X' axis being along the earthward normal to the shock and the Y' axis along the shock (see Figure 14). We assume for simplicity that the free stream solar wind velocity, the interplanetary magnetic field, and the shock normal all lie in the $X'Y'$ plane. In general, these three vectors are not coplanar; however, in the depletion layer analysis, only flux tubes that cross the shock near the stagnation streamline will be depleted significantly, and to a good approximation, $\mathbf{B}_{SW}$, $\mathbf{v}_{SW}$, and $\hat{X}'$ will be coplanar for these particular tubes. The solar wind magnetic field $\mathbf{B}_{SW}$ makes a preshock angle of ψ with the $\hat{Y}'$ axis, which implies

$$\begin{aligned} \mathbf{B}_{SW}\cdot\hat{X}' &= B_{SW}\sin\psi \qquad \mathbf{v}_{SW}\cdot\hat{X}' = v_{SW}\cos\psi \\ \mathbf{B}_{SW}\cdot\hat{Y}' &= -B_{SW}\cos\psi \qquad \mathbf{v}_{SW}\cdot\hat{Y}' = v_{SW}\sin\psi \end{aligned} \tag{D1}$$

Combining the jump conditions for a strong MHD shock with (D1) gives for $\gamma = \frac{5}{3}$,

$$\begin{aligned} \mathbf{B}(y, 0)\cdot\hat{X}' &= B_{SW}\sin\psi \\ \mathbf{v}(y, 0)\cdot\hat{X}' &= \tfrac{1}{4}v_{SW}\cos\psi \\ \mathbf{B}(y, 0)\cdot\hat{Y}' &= -4B_{SW}\cos\psi \\ \mathbf{v}(y, 0)\cdot\hat{Y}' &= v_{SW}\sin\psi \end{aligned} \tag{D2}$$

where the argument $(y, 0)$ refers to a distance y from the X-Z plane and time zero, i.e., just postshock. By symmetry, $\psi = 0$ when $y = 0$, so that

$$|\mathbf{B}_0| = 4B_{SW} \tag{D3}$$

$$|\mathbf{v}_0| = |v_{X0}| = v_{SW}/4 \tag{D4}$$

The velocities $\mathbf{v}(y, 0)$, $\mathbf{v}_0$, and $\mathbf{v}_{X0}$ are all measured in the rest frame of the magnetosphere. The bow shock obliquity velocity contribution to the initial value of v_y, measured in the frame drifting with the flux tube, is given by $[\mathbf{v}(y, 0) - \mathbf{v}_{CS}]\cdot\hat{b}_t$, where $\hat{b}_t = \mathbf{B}(y, 0)/|\mathbf{B}(y, 0)|$ and $\mathbf{v}_{CS}$ is the velocity relative to the magnetosphere of a local grid point in the xyz coordinate system. Bearing in mind that a grid point stays a fixed distance from the $y = 0$ plane as measured along a field line, we assume $\mathbf{v}_{CS}$ has magnitude $v_{SW}/4$ and is directed away from the sun. We thus obtain from (D2) the following expression for the initial y component of the kick velocity, valid for small ψ:

$$v_{\text{kick}}(y, 0) = -v_{SW}[(\tfrac{15}{16})\psi - \tfrac{1}{4}\theta_t] \tag{D5}$$

where θ_t is the angle between $\mathbf{B}(y, 0)$ and $\mathbf{B}_{SW}$ at the shock. Since (D1) and (D3) imply that for $|\psi| \ll 1$,

$$\theta_t \simeq 3\psi(y)/4 \tag{D6}$$

we have

$$v_{\text{kick}}(y, 0) \simeq -(\tfrac{3}{4})v_{SW}\psi(y) \tag{D7}$$

Correction for the noninertial reference frame. Zwan [1974] has shown that the effect of the noninertiality of the coordinate system on the momentum equation is to add a centrifugal-acceleration term $\rho\hat{y}\cdot\mathbf{a}_c$ to the left side of (2b), where $\mathbf{a}_c$ is the acceleration of a local coordinate point in relation to an inertial frame. The largest contribution to $\mathbf{a}_c$ comes from the fact that just postshock the flux tube is traveling at a velocity of approximately $v_{SW}/4$ away from the sun, but this antisolar velocity component drops to near zero by the time the flux tube gets close to the magnetopause. Since most of the sunward acceleration of the coordinate system occurs in the first few minutes after the flux tube crosses the shock, we approximate its effect by adding a term $-\int\hat{y}\cdot\mathbf{a}_c\,dt$ to the assumed initial y component of velocity,

$$v_{nt}(y, 0) \simeq -v_{SW}\bar{\theta}(y)/4 \tag{D8}$$

where $\bar{\theta}$ is an average small angle between the magnetosheath magnetic field and the initial solar wind magnetic field.

Combining (D7) and (D8) gives our initial velocity array of

$$v_y = v(y, 0) \simeq -v_{SW}[(\tfrac{3}{4})\psi(y) + \tfrac{1}{4}\bar{\theta}(y)] \tag{D9}$$

or, in dimensionless variables, using (5), (53), and (D4) gives

$$v^*(y^*, 0) \simeq -1.51\xi^{-1/2}[(\tfrac{3}{4})\psi(y^*) + \tfrac{1}{4}\bar{\theta}(y^*)] \tag{D10}$$

Matching the shock shapes of *Spreiter et al.* [1966], we get for small ψ,

$$\psi(y^*) \simeq -0.64y^* \tag{D11}$$

Lacking an accurate way to estimate $\bar{\theta}(y^*)$, we assume that most of the acceleration occurs when the field lines are relatively near the magnetopause, so that $\bar{\theta}(y^*)$ is approximately the angle between the magnetopause and $\mathbf{B}_{SW}$. For the magnetopause shape used by *Spreiter et al.* [1966] this angle is given approximately by

$$\bar{\theta}(y^*) \approx \theta_f \approx -1.09y^* \tag{D12}$$

Substituting (D11) and (D12) into (D10) yields for small $|y^*|$,

$$v^*(y^*, 0) \approx 1.14y^*\xi^{-1/2} \tag{D13}$$

The formula we have used for our initial velocity array closely approximates (D13) for $|y^*| \lesssim 0.5$ but allows for the fact that the bow shock angle ψ and magnetopause angle θ_f increase less than linearly with y^* for large $|y^*|$.

Initial velocity in the Z direction. The expression for S^* is, from (23),

$$S^* = 2|v_{X0}^*|(1 + v_{Z0}^*\tau_0^*/Z_0^*)^{-1} \tag{D14}$$

We rewrite v_{X0}^* using (53). Adapting the argument that led to (D1) and (D2) to the case where $\hat{X}'$ and $\hat{Y}'$ are in the X-Z plane perpendicular to $\mathbf{B}_{SW}$ leads to the following expression, valid for small $|\psi|$:

$$v_{Z0}^* \simeq -3v_{SW}\psi/(4C_{A0}) \simeq -1.13\xi^{-1/2}\psi \tag{D15}$$

where we have used (D4) and (53). Assuming that the shock is cylindrically symmetric, we have, in analogy to (D11) but for the X-Z plane, $\psi \simeq -0.64Z_0^*$. With the relation

$$\tau_0^* = \xi^{1/2}/\sigma \tag{D16}$$

which comes from (36), (D14) becomes

$$S^* = 0.75\xi^{-1/2}(1 + 0.724\sigma^{-1})^{-1} \tag{D17}$$

Values of S^* range from 0.84 for $\xi = 0.27$ and $\sigma = 1$ to 1.57 for $\xi = 0.091$ and $\sigma = 1.25$.

Effect of the initial conditions. The effect of the initial kick in the Z direction is to multiply layer thicknesses by approximately 0.6. From one calculation done with $v^*(y^*, 0) \approx 0$ we conclude that the initial kick in the y direction given by (D13),

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including both effects of bow shock obliquity and effects of noninertiality of the coordinate system, decreases layer thickness by a factor of ~3.

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